PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
Capstone Coursework Deliverable • Academic Clearance Automation Engine

Software Requirements Specification (SRS)

Academic Grievance, No-Dues & Clearance Automation Engine

Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the complete functionality of the **Academic Grievance, No-Dues & Clearance Automation Engine** for **Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh**.

The system replaces manual paper-based clearance circulars with an automated digital platform. It manages student clearance across campus departments, handles student grievances with automated escalation, and issues tamper-proof digital clearance certificates.

1.2 Scope of the System

The platform covers the following key operational areas:

1. **User Identity & Roles:** Supports PEC students using their 8-digit Student ID (SID, e.g., `22103042`) and department staff using their official designations (such as *Chief Warden*, *Librarian*, *HOD Labs*, *Accounts Officer*, and *Dean Academic Affairs*).
2. **Three-Gate Clearance Workflow:**
   - **Gate 1 (Parallel / Concurrent):** Central Library, PEC Hostels, and Gymkhana review student records simultaneously without blocking each other.
   - **Gate 2 (Sequential Prerequisite):** Departmental Labs can only review and clear students after all Gate 1 departments have approved.
   - **Gate 3 (Final Settlement):** The Accounts Branch performs fee reconciliation and authorizes the release of the Caution Money refund after Gates 1 and 2 are fully approved.
3. **Digital Signatures & Security:** Each department sign-off generates a secure HMAC-SHA256 digital signature seal. Final certificates include an immutable SHA-256 verification hash.
4. **Grievance Resolution with SLA:** Students can dispute incorrect dues. The system calculates a resolution deadline based on ticket priority and automatically escalates unresolved tickets to the Dean Academic Affairs (DAA).
5. **Public Certificate Verification:** Anyone (such as employers, universities, or visa officers) can verify the authenticity of an issued certificate online without needing an account.

1.3 Definitions, Acronyms, and Abbreviations

- **PEC:** Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh.
- **SID:** Student Identification Number (8-digit unique code, e.g., `22103042`).
- **DAA:** Dean Academic Affairs, Punjab Engineering College.
- **DSA:** Dean Student Affairs, Punjab Engineering College.
- **Central Library:** Dr. B.R. Ambedkar Central Library, PEC Chandigarh.
- **Hostel Council:** Administration for PEC residences (Shivalik, Kurukshetra, Himalaya, Aravali, Vindhya, KCH).
- **Gymkhana:** PEC Sports Complex, Gymnasium, Grounds, and Swimming Pool facility.
- **Caution Money:** Refundable security deposit (₹10,000) collected during admission and refunded upon final clearance.
- **CoR:** Chain of Responsibility Design Pattern (routes requests through approval handlers).
- **SLA:** Service Level Agreement (maximum time allowed to resolve a student grievance).
- **HMAC-SHA256:** Keyed-Hash Message Authentication Code (cryptographic signature for digital sign-offs).
- **INR (₹):** Indian Rupee, the official institutional currency.

1.4 References

- IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*.
- FIPS PUB 180-4: *Secure Hash Standard (SHS)*.
- RFC 2104: *HMAC: Keyed-Hashing for Message Authentication*.
- PEC Academic Ordinances & Regulations for B.Tech., M.Tech., and Ph.D. Degree Programs.
- AICTE / UGC Guidelines on Digital Grievance Redressal in Higher Educational Institutions.

1.5 Overview of the Document

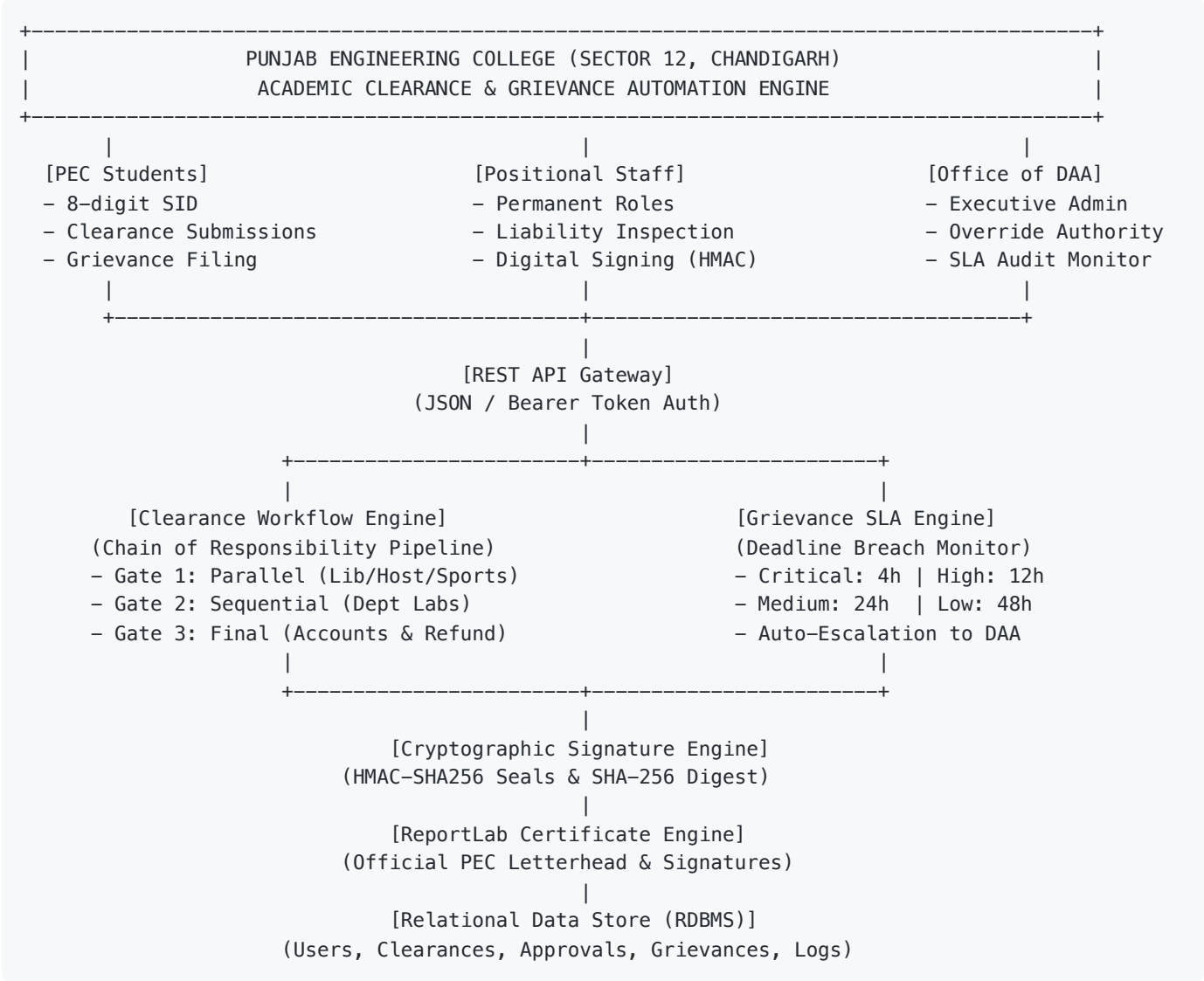
- **Section 2 (Overall Description):** Describes the general system architecture, high-level features, user roles, constraints, and operational assumptions.
- **Section 3 (Specific Requirements):** Detailed interface requirements, functional requirements divided into 6 modular features (`FR–1.1` to `FR–6.3`), performance requirements, and system quality attributes.
- **Section 4 (Appendix):** Quality compliance verification matrix verifying strict IEEE 830-1998 compliance.

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2. Overall Description

2.1 Product Perspective & Context

The system serves as the central digital clearance bridge between students, campus facilities, and the Office of the Dean Academic Affairs:



2.2 Product Functions (High-Level Summary)

The platform performs the following main functions:

1. **Login & Role Management:** Secure login with role-based access for Students, Staff, and Administrators.
2. **Clearance Request Tracking:** Visualizes clearance progress across Gate 1 (Library, Hostels, Sports), Gate 2 (Labs), and Gate 3 (Accounts).
3. **Department Review & Digital Signing:** Allows staff to check asset returns, record outstanding dues in INR (`₹`), and digitally sign approvals.
4. **Administrative Overrides:** Allows the Dean Academic Affairs to resolve blocked approvals with complete audit logs.
5. **Grievance Arbitration & Auto-Escalation:** Tracks student complaints and escalates them to the DAA if not resolved before the SLA deadline.
6. **Certificate Minting & PDF Download:** Generates an official signed PDF certificate with a verifiable QR/hash code once all 5 departments approve.

2.3 User Classes & Characteristics

1. **Student:**
   - Identified by an 8-digit PEC SID (e.g., `22103042`) and `@pec.edu.in` email.

- Submits a clearance application for milestones like Graduation or 6-Month Industrial Training.
- Views pending dues in INR (₹) and tracks real-time clearance status.
- Can file grievances if dues or remarks are disputed.

2. **Department Staff (Positional Roles):**
   - Accounts represent permanent institutional roles: *Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, and *Accounts Officer*.
   - Reviews student liabilities, marks approvals or rejections, and signs off with cryptographic HMAC seals.
3. **Dean Academic Affairs (DAA / Admin):**
   - Highest administrative authority at PEC.
   - Views institutional analytics, department resolution bottlenecks, and escalated grievances.
   - Has executive authority to override stalled approvals.
4. **Public Verifier:**
   - Outside organizations (employers, graduate schools, background checkers).
   - Can verify certificate validity online using the certificate ID or SHA-256 hash without logging in.

2.4 General Constraints

1. **PEC-Specific Constraints:**
   - Student IDs must follow the 8-digit format: `YYBBBRRR` (e.g., `22103042`).
   - All financial amounts must be in Indian Rupees (₹ / INR).
   - Staff accounts use official titles rather than individual personal names to remain stable across faculty rotations.
2. **Workflow Rules:**
   - Gate 2 (Department Labs) cannot be approved until all Gate 1 departments (Library, Hostels, Sports) are approved.
   - Gate 3 (Accounts) cannot be approved until Gates 1 and 2 are fully approved.
   - Final clearance certificate is only generated when all 5 departments reach `APPROVED` status.

2.5 Assumptions & Dependencies

1. Users access the web application through standard web browsers (Chrome, Edge, Firefox, Safari) on desktop or mobile.
2. The server clock is synchronized using Network Time Protocol (NTP) to calculate exact SLA deadlines.
3. Each student has at most one active clearance request at a time.

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3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Student Dashboard:** Clean responsive web interface showing the 3-gate progress bar, department status cards, outstanding dues in INR (₹), a button to submit grievances, and a download button for the clearance certificate.

- **Staff Console:** Review queue filtered by department where staff can inspect requests, enter dues in INR (`₹`), add remarks, and click "Approve & Sign" or "Reject".
- **DAA Admin Portal:** Executive dashboard showing campus clearance statistics, department SLA performance, overdue grievance alerts, and administrative override buttons.
- **Public Verification Page:** Simple public web page with an input box to search by Certificate ID or Hash and view instant verification results.

3.1.2 Hardware Interfaces

- **Server:** Runs on any standard 64-bit server or cloud VM (minimum 1 vCPU, 512 MB RAM).
- **Client Devices:** Any standard laptop, desktop, tablet, or smartphone.

3.1.3 Software Interfaces

- **Web Backend:** Python 3.12+ with FastAPI framework.
- **Database:** Relational database (SQLite for local execution, PostgreSQL for cloud deployment) managed via SQLAlchemy ORM.
- **PDF Engine:** ReportLab library for generating vector-quality PDF certificates.
- **Frontend Client:** React 19 Single Page Application with Tailwind CSS.

3.1.4 Communications Interfaces

- **Protocol:** HTTP/HTTPS using standard REST API communication with JSON payloads.
- **Authentication:** Stateless JSON Web Tokens (JWT) passed via standard `Authorization: Bearer <TOKEN>` HTTP headers.

3.2 Functional Requirements

3.2.1 Module 1: Authentication & User Accounts

- **FR-1.1 [Login Verification]:** The system shall authenticate users with institutional email and password. Passwords must be hashed using salted Bcrypt before storage.
- **FR-1.2 [Role-Based Access Control]:** The system shall restrict actions based on user role (`STUDENT`, `STAFF`, `ADMIN`). Students cannot access staff review queues; staff can only access their assigned department.
- **FR-1.3 [Positional Staff Titles]:** Staff profiles shall be linked to permanent department codes (`LIBRARY`, `HOSTEL`, `SPORTS`, `LABS`, `ACCOUNTS`) and official titles rather than individual faculty names.

3.2.2 Module 2: Clearance Request Management

- **FR-2.1 [Single Active Request]:** A student shall be allowed to submit a clearance request only if they have no other pending clearance request active.
- **FR-2.2 [Milestone Justification]:** Each clearance request must record the academic reason:
  - B.Tech Final Degree Graduation
  - Mandatory 6-Month Industrial Training / Internship (Semester 7 or 8)
  - M.Tech / Ph.D. Thesis Submission
  - Program Withdrawal / Caution Money Settlement

- **FR-2.3 [Automatic Gate Creation]:** When a student applies, the system shall automatically create 5 departmental clearance gate entries:
  - Central Library (Gate 1, status: `PENDING`)
  - PEC Hostels (Gate 1, status: `PENDING`)
  - Gymkhana & Sports (Gate 1, status: `PENDING`)
  - Department Labs (Gate 2, status: `WAITING_PREREQUISITE`)
  - Accounts Branch (Gate 3, status: `WAITING_PREREQUISITE`)

3.2.3 Module 3: Three-Gate Approval Pipeline (Chain of Responsibility)

- **FR-3.1 [Gate 1 Parallel Review]:** Central Library, Hostels, and Gymkhana shall review and approve/reject student records independently and concurrently.
- **FR-3.2 [Gate 2 Sequential Lab Prerequisite]:** Department Labs cannot approve clearance until Central Library, Hostels, and Gymkhana are all marked `APPROVED` (unless overridden by the DAA).
- **FR-3.3 [Gate 3 Accounts & Caution Money Release]:** The Accounts Branch cannot approve clearance until Gates 1 and 2 are fully approved. Upon approval, Accounts reconciles institutional dues and authorizes the release of the student's refundable Caution Money deposit (₹10,000).
- **FR-3.4 [Digital Signature Generation]:** When a department signs off on an approval, the system shall automatically compute a cryptographic HMAC-SHA256 digital signature:

  ```
  Digital Signature = HMAC-SHA256(Secret_Key, staff_id + dept_code + request_id + timestamp)
  ```

- **FR-3.5 [Final Certificate Minting]:** When all 5 gates reach `APPROVED` status, the overall clearance request shall transition to `APPROVED`. The system shall automatically generate a unique Certificate ID (format: `PEC-CLR-YYYY-XXXXX`) and an immutable SHA-256 certificate digest.

3.2.4 Module 4: Administrative Overrides & Audit Trail

- **FR-4.1 [DAA Executive Override]:** The Dean Academic Affairs shall have authority to override a rejected department approval, changing its status to `APPROVED` with reason notes.
- **FR-4.2 [Automatic Gate Unlocking]:** When an override is applied, the system shall automatically unlock downstream dependent gates (Gate 2 or Gate 3).
- **FR-4.3 [Immutable Audit Logging]:** Every approval, rejection, dues modification, grievance resolution, and administrative override shall be recorded in the system audit log with actor ID, action name, timestamp, and details.

3.2.5 Module 5: Student Grievances & Dynamic SLA Escalation

- **FR-5.1 [Filing a Grievance]:** A student can submit a grievance against any department reporting dues, providing subject, description, priority, and disputed amount in INR (`₹`).
- **FR-5.2 [Dynamic SLA Deadline]:** The system shall compute an exact deadline upon grievance submission based on priority:

  ```
  Resolution Deadline = Submission_Timestamp + Priority_Window
  - CRITICAL Priority: 4 Hours
  - HIGH Priority:     12 Hours
  - MEDIUM Priority:   24 Hours
  - LOW Priority:      48 Hours
  ```

- **FR-5.3 [Automated Escalation to DAA]:** If the current time exceeds the resolution deadline and the grievance is still `OPEN`, the system shall automatically mark it as `ESCALATED` (`is_escalated = True`) and notify the Dean Academic Affairs dashboard.
- **FR-5.4 [Grievance Resolution]:** The department staff or DAA can resolve a grievance by entering resolution notes and updating or waiving the associated dues in INR (`₹`).

3.2.6 Module 6: Certificate Generation & Public Verification

- **FR-6.1 [Cryptographic Hash Calculation]:** The certificate hash shall be computed as:

```
Certificate Hash = SHA-256(cert_id + student_sid + academic_session + completion_timestamp)
```

- **FR-6.2 [Official PDF Generation]:** The system shall generate a downloadable PDF certificate containing:
  - Official PEC letterhead, crest, and Sector 12 address.
  - Student details (Full Name, 8-digit SID, Degree, Branch).
  - Department clearance table showing timestamps, positional titles, and digital signatures.
  - Caution Money refund confirmation in INR (`₹`).
  - Cryptographic SHA-256 verification hash.
  - Signature blocks for Dean Academic Affairs (DAA) and Dean Student Affairs (DSA).
- **FR-6.3 [Public Online Verification]:** The system shall provide a public verification page `/api/v1/certificate/verify` where anyone can search by Certificate ID or Hash to verify authenticity without logging in.

3.3 Performance Requirements

- **NFR-1.1 [Fast Response Time]:** 95% of normal API requests (reading student progress, loading dashboards) shall respond in under 150 milliseconds.
- **NFR-1.2 [Instant PDF Generation]:** Official PDF clearance certificate generation shall complete in under 1.5 seconds.
- **NFR-1.3 [Quick SLA Evaluation]:** Automated evaluation of all active grievance deadlines shall execute in under 500 milliseconds.

3.4 Design Constraints

- **NFR-2.1 [Database Agnostic]:** All database operations must use SQLAlchemy ORM abstractions so the system can run seamlessly on both SQLite and PostgreSQL.
- **NFR-2.2 [Self-Contained Auth]:** The system implements self-contained JWT token authentication without requiring external single sign-on (SSO) infrastructure.
- **NFR-2.3 [SQL Injection Prevention]:** All queries must use parameterized ORM queries to eliminate SQL injection vulnerabilities.

3.5 Software System Attributes

3.5.1 Security

- **NFR-3.1 [Token Authentication]:** All protected endpoints require a valid JWT token signed with an institutional server secret.

- **NFR-3.2 [Cryptographic Sign-Offs]:** Each department approval is sealed with HMAC-SHA256. Any manual tampering with database records invalidates the signature seal.
- **NFR-3.3 [Student Privacy]:** Students can only view their own clearance records, dues, and grievances.

3.5.2 Concurrency & Data Consistency

- **NFR-4.1 [Concurrent Reviews]:** Simultaneous reviews by Gate 1 staff (Hostel, Library, Sports) execute independently without race conditions or lost updates.
- **NFR-4.2 [Live Screen Sync]:** Client web dashboards synchronize with the server every 8 to 10 seconds so status updates show up across student, staff, and admin screens without needing a page refresh.

3.5.3 Reliability & Availability

- **NFR-5.1 [Fault Isolation]:** A delay in one department (e.g. Sports complex staff reviewing records) will not block independent reviews in other Gate 1 departments.
- **NFR-5.2 [Transactional Integrity]:** Database updates (such as unblocking Gate 2 when Gate 1 completes) execute as atomic database transactions.

3.5.4 Maintainability

- **NFR-6.1 [Design Pattern Standard]:** The clearance routing engine strictly implements the Chain of Responsibility design pattern, allowing new approval departments to be added easily.
- **NFR-6.2 [PEP 8 Code Quality]:** All backend Python code adheres strictly to PEP 8 standards with 0 Flake8 violations.

3.6 Architectural Invariants (Permanently Frozen Rules)

1. **Three-Gate Structure:** The workflow structure (Gate 1 Parallel -> Gate 2 Sequential Labs -> Gate 3 Accounts Settlement) is permanently frozen.
2. **Positional Roles:** Staff roles are based on institutional administrative designations (DAA, Chief Warden, Librarian, Gymkhana President, HOD Labs, Accounts Officer).
3. **Institutional Currency:** All monetary amounts are in Indian Rupees (`₹` / INR).
4. **Student ID Format:** Student IDs are strictly 8-digit PEC SIDs (`YYBBBRRR`).
5. **Cryptographic Validation:** Issued certificates are validated using SHA-256 cryptographic digests.

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4. Appendix: IEEE Std 830-1998 Quality Compliance Matrix

This specification satisfies all 8 quality attributes specified in IEEE Std 830-1998:

IEEE 830 Quality Attribute	Fulfillment in this Specification	Section Reference	Verification Method
1. Correctness	Stated requirements accurately reflect Punjab Engineering College clearance regulations.	Sections 1.2, 3.2	Stakeholder Review
2. Unambiguity	Every requirement has a unique identifier (`FR-X.Y` , `NFR-X.Y`) with clear inputs, outputs, and formulas.	Section 3.2	Traceability Matrix
3. Completeness	Covers all campus clearance departments, fee settlements, grievance SLAs, and PDF certificate generation.	Sections 2.1, 3.2	Requirements Checklist
4. Consistency	Gate dependency rules, user permissions, and state transitions are mathematically non-contradictory.	Sections 3.2, 3.6	Automated Test Suite
5. Ranked for Importance	Three-gate pipeline and digital signatures are designated as **Essential & Frozen Invariants**.	Section 3.6	Priority Tagging
6. Verifiability	Every functional and non-functional requirement is verified through automated tests or API verification.	Sections 3.2, 3.3	Pytest Suite (11/11 Passed)
7. Modifiability	Organized into 6 clear modular subsections allowing easy navigation and maintenance.	Sections 3.1 - 3.2	Hierarchical Modular Layout
8. Traceability	Forward and backward traceability maintained from user personas to functional requirements and UML models.	Sections 2.3, 3.2	Model Cross-References